LAB EXPERIMENT RECORD

Set 29: Student Academic Performance

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	22 October 2026
Day	Thursday

Q1. Stacked Area Chart of Test Score and Participation Score

Aim: To create a stacked area chart showing test and participation scores across students.

R Code:

R

```
library(ggplot2)

data <- data.frame(

  Student_ID = c("S1", "S2", "S3", "S4", "S5"),

  Age = c(19, 21, 20, 22, 23),

  Study_Hours = c(12, 8, 15, 10, 7),

  Attendance = c(90, 70, 95, 85, 60),

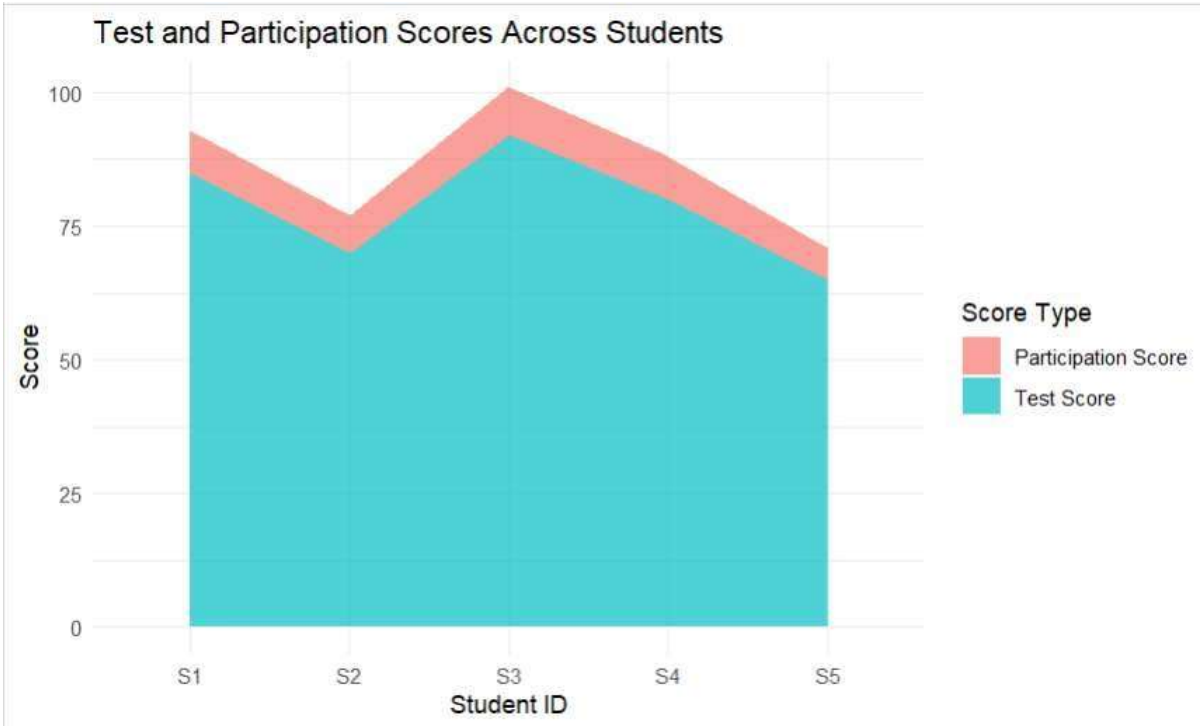
  Test_Score = c(85, 70, 92, 80, 65),

  Participation_Score = c(8, 7, 9, 8, 6)

)
```

```
area_data <- data.frame(
  Student_ID = rep(data$Student_ID, 2),
  Score_Type = rep(c("Test Score", "Participation Score"),
                   each = nrow(data)),
  Score = c(data$Test_Score, data$Participation_Score)
)

ggplot(area_data, aes(x = Student_ID, y = Score,
                      fill = Score_Type, group = Score_Type)) +
  geom_area(alpha = 0.7, position = "stack") +
  labs(title = "Test and Participation Scores Across Students",
       x = "Student ID", y = "Score",
       fill = "Score Type") +
  theme_minimal()
```



Observation: Student S3 has the highest test score (92) and participation score (9). Student S5 has the lowest test score (65) and participation score (6). Both scores generally follow a similar pattern across the five students.

Result: The stacked area chart compares test and participation scores across students.

Q2. Boxplot of Study Hours Grouped by Attendance Quartiles

Aim: To compare study hours among students grouped by attendance quartiles.

R Code:

R

```
library(ggplot2)

data <- data.frame(

  Student_ID = c("S1", "S2", "S3", "S4", "S5"),

  Study_Hours = c(12, 8, 15, 10, 7),

  Attendance = c(90, 70, 95, 85, 60)

)

data$Attendance_Quartile <- cut(

  data$Attendance,

  breaks = quantile(data$Attendance,

                    probs = seq(0, 1, 0.25)),

  include.lowest = TRUE,

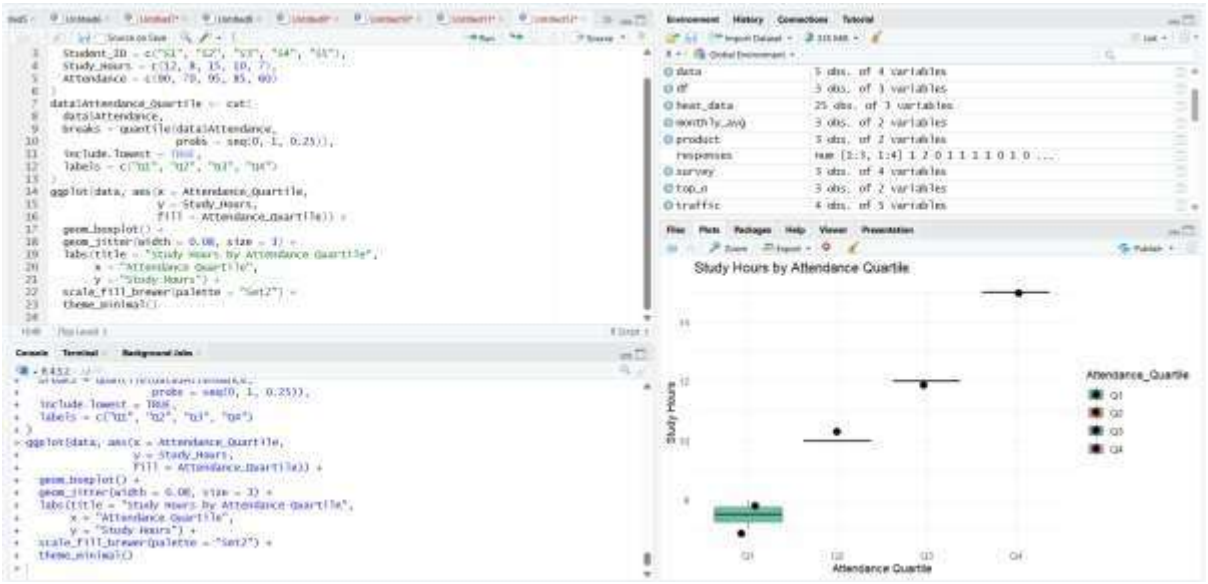
  labels = c("Q1", "Q2", "Q3", "Q4")

)

ggplot(data, aes(x = Attendance_Quartile,

                 y = Study_Hours,
```

```
            fill = Attendance_Quartile)) +
  geom_boxplot() +
  geom_jitter(width = 0.08, size = 3) +
  labs(title = "Study Hours by Attendance Quartile",
       x = "Attendance Quartile",
       y = "Study Hours") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```



Observation: Students with higher attendance generally report more study hours in this dataset. However, there are only five observations, so some quartiles contain very few students. Median and interquartile range estimates are therefore limited, and no strong outlier conclusion can be made.

Result: The boxplot compares study hours across attendance groups. Jittered points help show individual observations.

Q3. Density Plot of Test Scores

Aim: To examine the distribution of test scores and identify possible unusual values.

R Code:

R

```
library(ggplot2)

test_scores <- data.frame(

  Test_Score = c(85, 70, 92, 80, 65)

)

ggplot(test_scores, aes(x = Test_Score)) +

  geom_density(fill = "skyblue",

               color = "blue",

               alpha = 0.6,

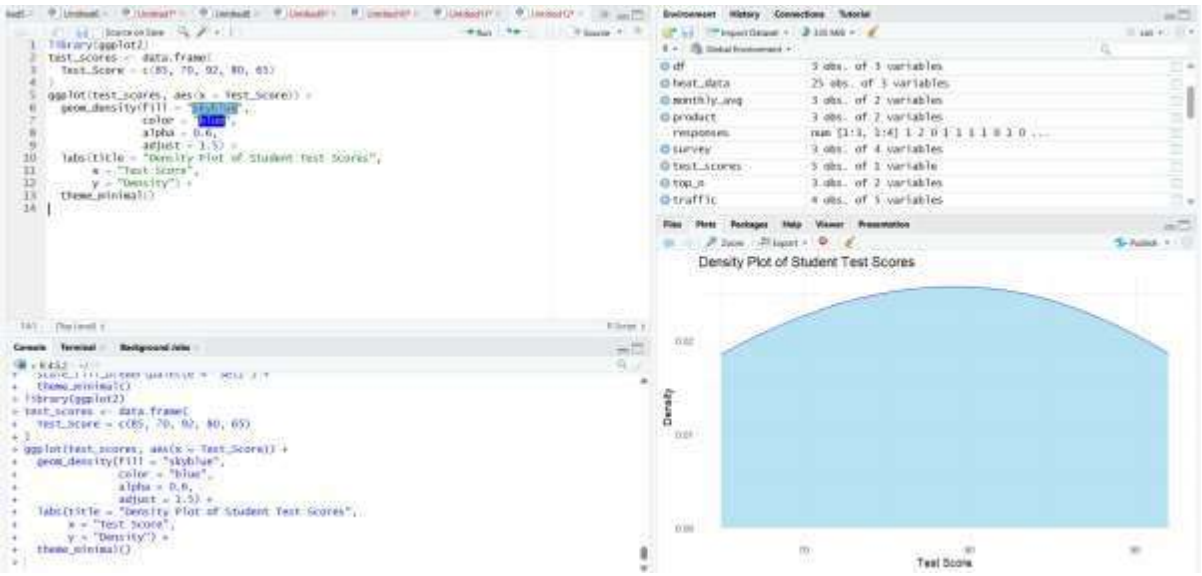
               adjust = 1.5) +

  labs(title = "Density Plot of Student Test Scores",

       x = "Test Score",

       y = "Density") +

  theme_minimal()
```



Observation: Test scores range from 65 to 92. The scores show variation among students, but with only five observations, the density curve is only a rough estimate. The plot alone cannot establish whether any score is an outlier.

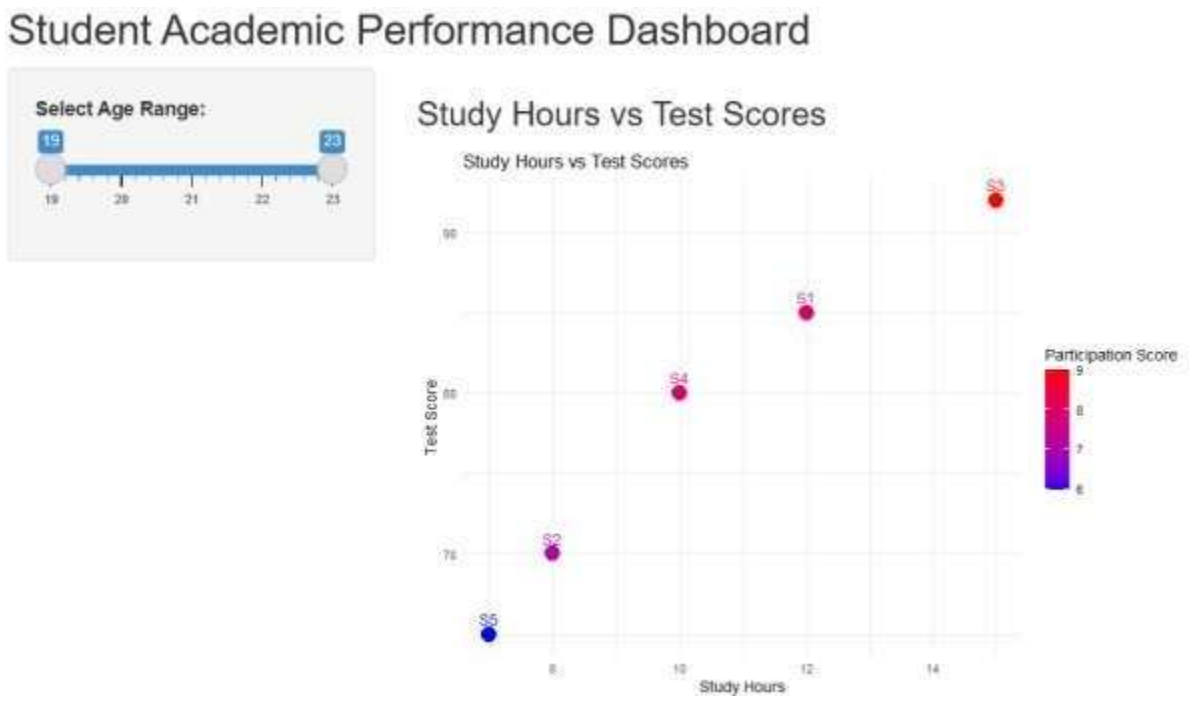
Result: The density plot visualizes the distribution of test scores.

Q4. Tableau Interactive Scatterplot

Aim: To create an interactive scatterplot showing the relationship between study hours and test scores, colored by participation score.

Procedure:

1. Enter the dataset in Excel and save it as Student_Academic_Performance.xlsx.
2. Open Tableau and connect to the Excel file.
3. Drag Study_Hours to **Columns** and Test_Score to **Rows**.
4. Drag Participation_Score to **Color** on the Marks card and Student_ID to **Label**.
5. Drag Age to **Filters**, select the required age values, and choose **Show Filter**.
6. Create a dashboard and add the scatterplot. Configure tooltips to show Student ID, age, study hours, attendance, test score, and participation score.



Filtered Student Details

Student_ID	Age	Study_Hours	Attendance	Test_Score	Participation_Score
S1	19.00	12.00	90.00	85.00	8.00
S2	21.00	8.00	70.00	70.00	7.00
S3	20.00	15.00	95.00	92.00	9.00
S4	22.00	10.00	85.00	80.00	8.00
S5	23.00	7.00	60.00	65.00	6.00

Analytical Interpretation:

- S3 studies 15 hours and scores 92, the highest test score.
- S5 studies 7 hours and scores 65, the lowest test score.
- S1 studies 12 hours and scores 85, showing a relatively high score.
- Participation scores range from 6 to 9 and can be compared with academic performance using color.
- The age filter allows users to compare students of different ages.

Result: The interactive Tableau scatterplot helps explore the relationship between study hours, test scores, and participation while filtering students by age.